Cyclic Codes

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Cyclic Codes

Basics of Cyclic Code

- Cyclic Code is a Type of Linear Block Code with Cyclic Properties.

Properties of Cyclic Code

1. Linearity Property
   - If we have two Cyclic Codewords C_i and C_j, then their sum $C_p = C_i + C_j$ is also a Codeword.
2. Cyclic Property
   - After performing a cyclic shift on any codewor

❑ Example 1: [0000, 0110, 1001, 1111], Is it a Cyclic Code?

~ Linearity

$$
\begin{array}{r}
0110 \\
-\;1001 \\
\hline
1111
\end{array}
$$

← Valid Codeword

~ Cyclic

$\boxed{0110}$

0011 ← Not a Valid Codeword

~ So, It is not Cyclic Code

Example 2: [0000, 0101, 1010, 1111], Is it a Cyclic Code?

- linearity.

$$\begin{array}{r} 0101 \\ 1010 \\ \hline 1111 \end{array}$$

← Valid Codeword

- Cyclic

0101

1010 ← Valid Codeword

- So, It is cyclic code.

Generator Polynomial

In a **cyclic code**, every codeword is divisible by a special polynomial called the **generator polynomial**, denoted by:

$$g(x)$$

If $c(x)$ is a valid codeword, then:

$$c(x) \mod g(x) = 0$$

That means:

$$c(x) = m(x) \cdot g(x)$$

where:

- $m(x)$ = message polynomial
- $g(x)$ = generator polynomial
- $c(x)$ = codeword polynomial

Source: chatgpt

Generator Polynomial

For a binary (n, k) cyclic code:

1. $g(x)$ must divide:

$$x^n - 1$$

   over GF(2)

2. Degree of $g(x) = n - k$
3. Coefficients are in GF(2) (0 or 1)

Example: (7,4) Cyclic Code

We want:

- $n = 7$
- $k = 4$
- So degree of $g(x) = 3$

Now factor:

$$x^7 - 1 = (x + 1)(x^3 + x + 1)(x^3 + x^2 + 1)$$

Possible generator polynomials:

- $g(x) = x^3 + x + 1$
- $g(x) = x^3 + x^2 + 1$

Both define different (7,4) cyclic codes.

Source: chatgpt

Message Polynomial

Suppose a message has **k bits**:

$$m = (m_{k-1}, m_{k-2}, \dots, m_1, m_0)$$

Each $m_i \in \{0, 1\}$.

The corresponding **message polynomial** is:

$$m(x) = m_{k-1}x^{k-1} + m_{k-2}x^{k-2} + \dots + m_1 x + m_0$$

So:

- Each bit becomes a coefficient.
- The position of the bit determines the power of x.
- Operations are usually performed **modulo 2** (binary arithmetic).

Let:

$$m = (1, 1, 0, 0, 1)$$

Then:

$$m(x) = 1x^4 + 1x^3 + 0x^2 + 0x + 1$$

$$m(x) = x^4 + x^3 + 1$$

Source: chatgpt

Non-Systematic Cyclic Code

- Example 1: Find code vector for the Non-Systematic Cyclic Code (7, 4) with the Generator polynomial $G(P) = P^3 + P + 1$ and the Message M = [1 0 1 1]

(n, k)

└ n = 7

k = 4

~ Systematic - $\underbrace{x\,x\,x\,x}_{\text{Info}}$ $\underbrace{x\,x\,x}_{\text{Parity}}$

~ $G(P) = P^3 + P + 1$

$$M = \begin{array}{cccc} & & & \\ [1 & 0 & 1 & 1] \\ P^3 & P^2 & P^1 & P^0 \end{array}$$

$m(P) = P^3 + P + 1$

~ $C(P) = m(P)\,G(P)$

$= (P^3 + P + 1)(P^3 + P + 1)$

$= P^6 + \cancel{P^4} + \cancel{P^3} + \cancel{P^4} + P^2 + \cancel{P} + \cancel{P^3} + \cancel{P} + 1$

$= P^6 + P^2 + 1$

$$\sim C = \begin{array}{ccccccc} [1 & 0 & 0 & 0 & 1 & 0 & 1] \\ P^6 & P^5 & P^4 & P^3 & P^2 & P^1 & P^0 \end{array}$$

- Example 2: Find code vector for the Non-Systematic Cyclic Code (7, 4) with the Generator polynomial $G(P) = P^3 + P^2 + 1$ and the Message M = [1 0 1 0]

$n = 7$

$k = 4$

$G(P) = P^3 + P^2 + 1$

$$M = \begin{array}{cccc} & & & \\ \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 1 & 0 \\ P^3 & P^2 & P^1 & P^0 \end{array}$$

$M = [1\ 0\ 1\ 0]$

$m(P) = P^3 + P$

$C(P) = m(P)\,G(P)$

$= (P^3 + P)(P^3 + P^2 + 1)$

$= P^6 + P^5 + \cancel{P^3} + P^4 + \cancel{P^3} + P$

$= P^6 + P^5 + P^4 + P$

$$C = \begin{array}{ccccccc} P^6 & P^5 & P^4 & P^3 & P^2 & P^1 & P^0 \\ \end{array}$$

$C = [1\ 1\ 1\ 0\ 0\ 1\ 0]$

$$\begin{array}{ccccccc} P^6 & P^5 & P^4 & P^3 & P^2 & P^1 & P^0 \end{array}$$

$$G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Systematic Cyclic Code

Basics of Systematic Cyclic Code

- Format of Systematic Cyclic Code (n, k) is given by:

$$\Longrightarrow \textbf{Codeword} = [\textbf{Message bits (k bits), Parity bits (n − k bits)}]$$

- Codeword of Systematic Cyclic Code is given by:

$$\Longrightarrow C(x) = x^{n-k}m(x) + p(x)$$

  - C(x) = Codeword Polynomial
  - m(x) = Message Polynomial
  - p(x) = Parity Polynomial

- Parity Polynomial p(x) of Systematic Cyclic Code is given by:

$$\Longrightarrow p(x) = \text{Rem}\left[\frac{x^{n-k}m(x)}{g(x)}\right]$$

  - g(x) = Generator Polynomial

- Example 1: Find code vector for the Systematic Cyclic Code (7, 4) with the Generator Polynomial $G(x) = x^3 + x^2 + 1$ and the message [1 0 1 0]

$\rightarrow (7,4) = (n,k)$

$- n = 7$

$k = 4$

$\rightarrow G(x) = x^3 + x^2 + 1$

$m = [1 \quad 0 \quad 1 \quad 0]$

$\quad\; x^3 \; x^2 \; x^1 \; x^0$

$m(x) = x^3 + x$

$\rightarrow c(x) = x^{n-k} m(x) + P(x)$

$\Rightarrow P(x) = \text{Rem}\left[\dfrac{x^3(x^3+x)}{x^3+x^2+1}\right]$

$= \text{Rem}\left[\dfrac{x^6+x^4}{x^3+x^2+1}\right]$

$$
\begin{array}{r|l}
 & x^3 + x^2 + 1 \\ \hline
x^3+x^2+1 & \cancel{x^6} + x^4 \\
 & \cancel{x^6} + x^5 + x^3 \\ \hline
 & \cancel{x^5} + \cancel{x^4} + x^3 \\
 & \cancel{x^5} + \cancel{x^4} + x^2 \\ \hline
 & \cancel{x^3} + \cancel{x^2} \\
 & \cancel{x^3} + \cancel{x^2} + 1 \\ \hline
 & \boxed{1}
\end{array}
$$

Rem

$\rightarrow c(x) = x^3(x^3+x) + 1$

$= x^6 + x^4 + 1$

Message Parity

$$
\rightarrow c = \begin{array}{c} [\,1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 1\,] \\ x^6 \;\; x^5 \;\; x^4 \;\; x^3 \;\; x^2 \;\; x^1 \;\; x^0 \end{array}
$$

Generator Matrix of Systematic Cyclic Code

Basics of Generator Matrix of Systematic Cyclic Code

- Generator Matrix G:

$$\Longrightarrow \quad \mathbf{G = [Identity\ I\ (k\ bits),\ Parity\ P\ (n-k\ bits)]}$$

- For Cyclic Code (n, k), then the Generator matrix G has n columns and k rows.
  - I = Identity is k x k Matrix
  - P = Parity is k x (n-k) Matrix
- Row of Parity Matrix can be calculated by:

$$\Longrightarrow \mathbf{1st\ Row = Rem}\left[\frac{x^{n-1}}{g(x)}\right]$$

$$\Longrightarrow \mathbf{2nd\ Row = Rem}\left[\frac{x^{n-2}}{g(x)}\right]$$

....

$$\Longrightarrow \mathbf{kth\ Row = Rem}\left[\frac{x^{n-k}}{g(x)}\right]$$

- **Example 1: The Generator Polynomial of Cyclic Code (7, 4) is given by $G(x) = x^3 + x + 1$. Then Construct Generator Matrix.**

~ $(7,4) = (n,k)$

- $n = 7$
- $k = 4$

$(k \times k)$ Identity

$(k \times (n-k))$ Parity

~ $G = [I : P] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$

~ 1^{st} Row $= \text{Rem}\left[\frac{x^{n-1}}{g(x)}\right] = \text{Rem}\left[\frac{x^6}{x^3+x+1}\right] = x^2+1 \quad [1\ 0\ 1]$

~ 2^{nd} Row $= \text{Rem}\left[\frac{x^{n-2}}{g(x)}\right] = \text{Rem}\left[\frac{x^5}{x^3+x+1}\right] = x^2+x+1 \quad [1\ 1\ 1]$

~ 3^{rd} Row $= \text{Rem}\left[\frac{x^{n-3}}{g(x)}\right] = \text{Rem}\left[\frac{x^4}{x^3+x+1}\right] = x^2+x \quad [1\ 1\ 0]$

$\rightsquigarrow 4^{th}$ Row $= \text{Rem}\left[\dfrac{x^{n-4}}{g(x)}\right] = \text{Rem}\left[\dfrac{x^3}{x^3+x+1}\right] = x+1$ $\quad [\,0\ \ 1\ \ 1\,]$

$$x^3+x+1 \,\overline{\big)\ \cancel{x^6}\ }\quad \text{quotient: } x^3+x+1$$

$$\begin{array}{l}
\cancel{x^6}+x^4+x^3 \\
\hline
\cancel{x^4}+x^3 \\
\cancel{x^4}+x^2+x \\
\hline
\cancel{x^3}+x^2+\cancel{x} \\
\cancel{x^3}+\cancel{x}+1 \\
\hline
x^2+1
\end{array}$$

$$x^3+x+1 \,\overline{\big)\ \cancel{x^5}\ }\quad \text{quotient: } x^2+1$$

$$\begin{array}{l}
\cancel{x^5}+x^3+x^2 \\
\hline
\cancel{x^3}+x^2 \\
\cancel{x^3}+x+1 \\
\hline
x^2+x+1
\end{array}$$

$$x^3+x+1 \,\overline{\big)\ \cancel{x^4}\ }\quad \text{quotient: } x$$

$$\begin{array}{l}
\cancel{x^4}+x^2+x \\
\hline
x^2+x
\end{array}$$

$$x^3+x+1 \,\overline{\big)\ \cancel{x^3}\ }\quad \text{quotient: } 1$$

$$\begin{array}{l}
\cancel{x^3}+x+1 \\
\hline
x+1
\end{array}$$

Designing of Cyclic Encoder

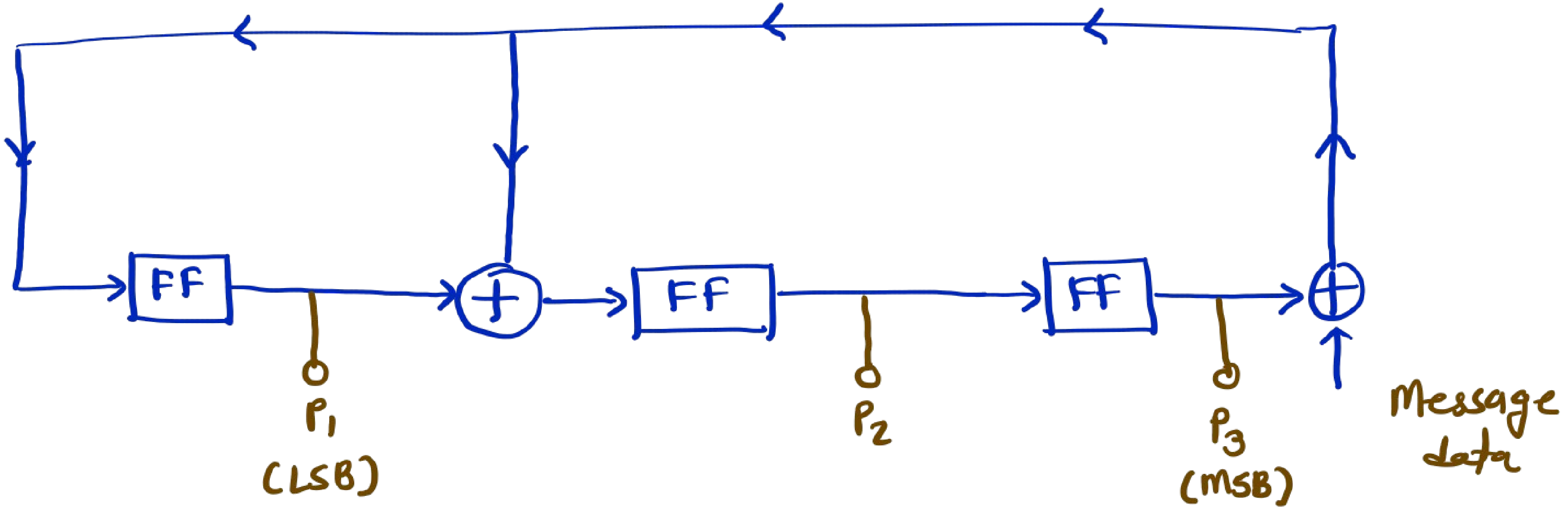
- Example 1: The Generator Polynomial of Cyclic Code (7, 4) is given by $G(x) = x^3 + x + 1$.

A. Design Cyclic Encoder Circuit/Hardware

B. Find Codeword if Message is [1 1 1 0]

$\rightsquigarrow$ $G(x) = 1 + x + x^3$

$= 1.x^0 + 1.x^1 + 0.x^2 + 1.x^3$



m	P_1	P_2	P_3
	0	0	0
1	1	1	0
1	1	0	1
1	0	1	0
0	0	0	1

$\rightsquigarrow$ Codeword = [Message , Parity]

= [1110 100]

CRC (Linear Feedback Shift Register) Algorithm

CRC (LFSR) Algorithm – Steps

1. **Initialize**
   - Let generator polynomial degree = `n`
   - Initialize shift register with `n` zeros
2. **Append Zeros**
   - Append `n` zeros to the input data
3. **Process Each Bit**
   - For each bit in the extended data:
     1. Compute `feedback = input_bit XOR leftmost_register_bit`
     2. Shift register left by 1 bit
     3. If `feedback = 1`, XOR register with generator polynomial (excluding MSB)
4. **Final Remainder**
   - After all bits are processed, contents of register = **CRC**
5. **Transmission**
   - Append CRC to original data
6. **Receiver Check**
   - Perform same division on received frame
   - If remainder = all zeros → **No error**
   - Else → **Error detected**

Source: chatgpt

Complete Example of Cyclic Code

- Example: The Generator Polynomial of the Systematic Cyclic Code (7, 4) is given by $G(x) = x^3 + x + 1$. Calculate the following Parameters:

A. Generator Matrix
B. Codeword for Message [1 0 1 1]
C. Parity Check Matrix
D. Decoding Matrix
E. Syndrome Matrix if received codeword is [1 1 0 1 1 0 0]
F. Error Correction with above received data

$\rightsquigarrow (7,4) = (n, k)$

- $n = 7$
- $k = 4$

$\rightsquigarrow G(x) = x^3 + x + 1$

$$G = [\, I_K : P \,] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Identity Parity

1st Row $= \text{Rem}\left[\dfrac{x^{n-1}}{g(x)}\right] = \text{Rem}\left[\dfrac{x^6}{x^3+x+1}\right] = x^2+1 \quad [1\ 0\ 1]$

2nd Row $= \text{Rem}\left[\dfrac{x^{n-2}}{g(x)}\right] = \text{Rem}\left[\dfrac{x^5}{x^3+x+1}\right] = x^2+x+1 \quad [1\ 1\ 1]$

3rd Row $= \text{Rem}\left[\dfrac{x^{n-3}}{g(x)}\right] = \text{Rem}\left[\dfrac{x^4}{x^3+x+1}\right] = x^2+x \quad [1\ 1\ 0]$

4th Row $= \text{Rem}\left[\dfrac{x^{n-4}}{g(x)}\right] = \text{Rem}\left[\dfrac{x^3}{x^3+x+1}\right] = x+1 \quad [0\ 1\ 1]$

$$
\begin{array}{r|l}
 & x^3+x+1 \\
\hline
x^3+x+1 & \cancel{x^6} \\
 & \cancel{x^6}+x^4+x^3 \\
\hline
 & \cancel{x^4}+x^3 \\
 & \cancel{x^4}+x^2+x \\
\hline
 & \cancel{x^3}+x^2+\cancel{x} \\
 & \cancel{x^3}+\cancel{x}+1 \\
\hline
 & x^2+1
\end{array}
\qquad
\begin{array}{r|l}
 & x^2+1 \\
\hline
x^3+x+1 & \cancel{x^5} \\
 & \cancel{x^5}+x^3+x^2 \\
\hline
 & \cancel{x^3}+x^2 \\
 & \cancel{x^3}+x+1 \\
\hline
 & x^2+x+1
\end{array}
\qquad
\begin{array}{r|l}
 & x \\
\hline
x^3+x+1 & \cancel{x^4} \\
 & \cancel{x^4}+x^2+x \\
\hline
 & x^2+x
\end{array}
\qquad
\begin{array}{r|l}
 & 1 \\
\hline
x^3+x+1 & \cancel{x^3} \\
 & \cancel{x^3}+x+1 \\
\hline
 & x+1
\end{array}
$$

$$\rightsquigarrow \; C = MG = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \qquad \rightsquigarrow \; P^T = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

~ Parity Check Matrix H $= [P^T : I_{n-k}] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

~ Decoding Matrix H^T $= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ← 1st bit error ← 2nd bit error ⋮ ← 7th bit error

0 0 0 ← No error

~ Syndrome S $= \text{Rem}\left[\dfrac{y(x)}{g(x)}\right] = \text{Rem}\left[\dfrac{x^6+x^5+x^3+x^2}{x^3+x+1}\right] = x^2+1$

[1 0 1]

$$\begin{array}{cccccccc} & x^6 & x^5 & x^4 & x^3 & x^2 & x^1 & x^0 \\ \rightsquigarrow\ y = & [\,1 & 1 & 0 & 1 & 1 & 0 & 0\,] \end{array}$$

$$y(x) = x^6 + x^5 + x^3 + x^2$$

$$g(x) = x^3 + x + 1$$

$$\begin{array}{r|l} & x^3 + x^2 + x + 1 \\ \hline x^3 + x + 1 & \cancel{x^6} + x^5 + \cancel{x^3} + x^2 \\ & \cancel{x^6} + x^4 + \cancel{x^3} \\ \hline & \cancel{x^5} + x^4 + \cancel{x^2} \\ & \cancel{x^5} + x^3 + \cancel{x^2} \\ \hline & \cancel{x^4} + x^3 \\ & \cancel{x^4} + x^2 + x \\ \hline & \cancel{x^3} + x^2 + \cancel{x} \\ & \cancel{x^3} + \cancel{x} + 1 \\ \hline & x^2 + 1 \end{array}$$

$$\rightsquigarrow\ e = [1\ 0\ 0\ 0\ 0\ 0\ 0]$$

$$y = [1\ 1\ 0\ 1\ 1\ 0\ 0]$$

$$\rightsquigarrow\ x = e \oplus y$$

$$= [0\ 1\ 0\ 1\ 1\ 0\ 0]$$

Correction

Cyclic Redundancy Check - CRC

❖ Basics of Cyclic Redundancy Check

- CRC uses a Divisor with L bits. {Divisor may be given in Galois fields.}
- Here we append L – 1 bits with data bits.
- Here we perform binary division for CRC generation.
- Receiver and Transmitter should be agreed upon by the same Divisor.

❖ Example of Cyclic Redundancy Check

- Data = 01010001, Divisor = 1001. Find transmitted codeword using CRC.

```
01010001000
 1001
 001100
   1001
   01011
    1001
    001000
      1001
      00010  ← CRC
```

Tx Codeword = 01010001 010

data: 01010001, crc: 010

❖ Example of Cyclic Redundancy Check

- Data = 1110010101, Divisor = $X^3 + X^2 + 1$. Find transmitted codeword using CRC. = 1101

```
1110010101 | 000
1101
--------
001101 01
  1101
-----------
00000101 | 0
     1101| 1
     --------
     011 | 10
      11 | 01
     --------
      00 | 110  ← CRC
```

Tx Codeword = $\underbrace{1110010101}_{\text{data}}\ \underbrace{110}_{\text{CRC}}$

Examples on CRC – Cyclic Redundancy Check

Example 1 – The message 11001001 is to be transmitted using the CRC polynomial $X^3 + 1$ to protect it from errors. The message that should be transmitted is ……………………. If the received codeword at the receiver is 11011001011 then how the receiver will take decision?

(~) Divisor = $x^3 + 1$ = 1001 ← L = 4 bits

```
1 1 0 0 1 0 0 1 | 0 0 0
1 0 0 1
-------
0 1 0 1 1
  1 0 0 1
  -------
  0 0 1 0 0 0
      1 0 0 1
      -----------
      0 0 0 1 1 | 0 0
            1 0 | 0 1
            ---------
            0 1 | 0 1 0
              1 | 0 0 1
              ---------
              0 | 0 1 1  → CRC
```

$Tx = \underbrace{11001001}_{data}\ \underbrace{011}_{CRC}$

$\rightarrow$ 1 1 0 1 1 0 0 1 0 1 1 , $D = 1001$

```
1101100 1|011
1001
----
01001
 1001
------------
 0000001|011
       1|001
       -----
       0|010
```

$\boxed{010} \leftarrow R_x \neq 0$

Error